Proceedings of the 4th International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT-2026)
IEEE Xplore Part Number: CFP26CV1-ART; ISBN: 979-8-3315-9654-5

A Novel Optimizer based Deep Neural Network Model for IoMT

Buditi Sriveni
Dept. of CSE, Malla Reddy University
Hyderabad, 500100
mailforsriveni@gmail.com

Naveen Kumar Navuri
Dept. of CSE, Malla Reddy University
Hyderabad, 500100
naveennavuri@gmail.com

Korra Srinivas
Dept. of CSE, AIML, Malla Reddy Engineering College for Women
Maisammaguda, Dhulapally,
Secunderabad, Telengana, 500100
ksrinivas023@gmail.com

Tirumala Paruchuri
Dept. of CSE, Malla Reddy Deemed to be University
Maisammaguda, Medchal,
Malkajgrir, Telengana, 500100
tirumalaparuchuri@gmail.com

Jaganathan Muthusamy
Dept. of CSE, Malla Reddy Viswavidyapeeth
Venkatarama Colony, Suraram,
Telengana, 500055
jaganathanm@bitsathy.ac.in

***Abstract*—Intelligent diagnostics and real-time monitoring in the field of Internet of Things (IoMT) is changing modern healthcare, which also raises the risk of cyber threats, which makes important to get a strong intrusion detection systems (IDS). The current research used the MSCAD dataset to show the efficiency of eight popular optimizer for training the deep neural network model, by considering the real IoMT network conditions. Analyzing different performance metrices shows that the DNN optimized with Adam performs the best as compared to others, achieving a precision of 0.97753. This level of accuracy shows just how important it is to choose the right optimizer when building effective deep learning-based intrusion detection systems for medical networks. The proposed DNN-Adam method provides useful information for creating secure and successful cybersecurity solutions for limited resources, real-time healthcare systems by enhancing detection accuracy, reliability and flexibility in the IoMT environment.**

Keywords—IoT, Deep Nural Network, IoMT, Machine Learning

I. Introduction

Many research shows that the Internet of Things (IoT) is the next major technological advancement after the Internet. Nowadays all the basic facilities including agricultural sector, healthcare practices, and disaster detection are only a few of its applications; its continuous operation needs a number of sensors, computing units and storage devices. However, IoT is a seamless connectivity; it still needs to deal with different issues like heterogeneous devices, diverse protocols, energy limitations and computational constraints, which lead to several security gaps that must be addressed to ensure protection [1]. Because many IoT devices have little security, they are at risk of problems with confidentiality, integrity, and authenticity. A great deal of security incidents are caused by unauthorised data or service loss, and physical security risks can be greatly increased by device-level vulnerabilities. To preserve the user privacy, it is essential to give the volume of personal data shared across different platforms with a secure mechanism will help to protect sensitive information and stop network intrusions [2]. As the population of the world is raising, the cost of Healthcare costs is rising along with the world's population, requiring the development of intelligent systems that can provide real-time medical insights.

To send medical data for communication through network, the Internet of Medical Things (IoMT) [3] been used which is the extension version of the Internet of Things, and data like blood pressure, temperature, and heart rate been monitored which help to take important measures to save life of patients. Even IoMT makes it possible to provide services in remote areas across the globe, by which patients get an immediate solution to their health problems. IoMT is depends on cloud, virtual communication tools, and extensive sensing Healthcare systems is depending on cloud computing, virtual communication tools and sophisticated sensing equipment in future, and it improves the complexity of the systems and helps to widen the scope of care. With these advantages, make the system worries about security and privacy continue to prevent cloud-based healthcare systems from becoming widely used. Traditional Intrusion Detection Systems (IDSs) are becoming less effective in this environment, and research on creating effective countermeasures to the threats posed by these developments is conspicuously lacking.

Sorting through medical data can be assisted by machine learning (ML). ML used different methods to find patterns in big datasets, which enhances healthcare operations like record management, precise forecasting, and patient treatment plan customisation [4-6]. With it, technology can reduce human error. There are certain challenges with traditional machine learning techniques, such as high processing costs, memory problems, and reduced accuracy when dealing with extremely complicated data. However, deep learning (DL), a simpler kind of machine learning, tackles these problems by capturing hierarchical representations using deep neural networks. Optimisation methods that adjust weights and biases to lower mistakes and also increase prediction accuracy which is most important to success of training these models [7].

Here, a deep neural network (DNN) model is designed whos work is to detect cyberattacks in healthcare networks. Several optimizers which including Adam, SGD, Adadelta, Adagrad, Nadam, Adamax, AdamW and RMSprop are evaluated to determine the most efficient option. Experimental findings show that the Adam-based DNN framework achieves superior accuracy and performance compared to traditional ML methods. The MSCAD (Multi-Step Cyber-Attack Dataset for Intrusion Detection) dataset is used to overcome limitations linked to medical network data of high dimensionality, with the primary goal of accurately identifying malicious activities.

In this research work which is organised as follows: Section 2 presents the outline, Section 3 gives a shorter analysis on early developed models, Section 4 summarises the essential background, Section 5 outlines the proposed methodology, Section 6 illustrates the experimental process, and Section 7 concludes the study.

II. RESEARCH OUTLINE

In past years, the IoMT has become more forward-thinking in medical sectors, which includes taking next level in the field of remote monitoring of healthcare services, playing a key role for proper identify the medical conditions of patients. As the requirement is rising, different interconnected healthcare devices used in IoMT display some security limitations, making them easy target as significant threat, as they can possibly control or manipulate the medical data with equipment, which may create issues towards lives of patients. For this reason, addressing these problems becomes more important to improve the performance and efficacy of medical IoT systems. At the same time, the IoMT system handles sensitive medical information, which is are attractive target for very sensitive data attacks like Web Crawling. In this type of attacks which are specifically target main safety elements, which including more confidentiality. And such attacks have the potential to influence the trust on the system and wide operation of IoMT models. These concerns motivated the present study, which aims to identify the major security risks affecting IoMT systems and to recommend appropriate strategies to strengthen their protection.

III. LITERATURE REVIEW

Recent studies have used machine learning to strengthen security within the IoMT ecosystem. Authors in [9] analysed NSL-KDD attacks mechanism targeting IoT sensor networks and compared over eleven ML models, finding that tree-based and ensemble methods performed best. XGBoost mechanism has remained the highest accuracy of 97% with 90.5% MCC and 99.6% AUC. Authors in [10] addressed six ML techniques for detecting attacks in MQTT-based communication. They tested packet-level and flow-based features using simulated MQTT traffic, showing that flow features are essential for identifying MQTT-specific threats, while packet features work well for general attacks.

Ahmad et al. [11] used a cluster-based intelligent engineering method for addressing the major issues like overfitting and data imbalance. Using RF, SVM and ANN, they reported 98.67% accuracy for binary classification and 97.37% for multiclass tasks. Iwendi et al. [12] developed a feature-optimized IDS by the hybridization of Random Forest and genetic algorithm. Their approach, evaluated on NSL-KDD using RF, NB and LR models, improved detection accuracy, recall and F1-score by selecting the most effective feature subsets, as mentioned in Table 1.

TABLE 1. A SUMMARY OF VARIOUS DEVELOPED METHODS

Dataset	Methods	Proposed Model	Optimizer Used	Performance Measure(%)	Ref.
Synthetic Dataset	ANN, DT, RF, KNN	ANN	Adam	91.01	[8]
NSL-KDD	XGBoost, Tree Based Methods (NB)	XGBoost	NA	97.02	[9]
MQTT Dataset	SVM, Decision Tree	RF	NA	99.046	[10]
UNSW-NB15	RF, SVM, DNN	ANN	Adam	96.964	[11]
NSL-KDD, CSE-CIC-IDS2018	RF, GA, NB, LR	RF	NA	98.8134	[12]

An advanced intrusion detection system for smart healthcare systems strengthens data processing and also enhances overall security, which also lowers the risk of important failures within IoMT networks. Numerous studies addressing different privacy concerns and cyberthreats in medical settings contribute to IoMT security. The simple solutions are typically more appropriate for real world healthcare applications, in other hand more complicated models may require more resources. In earlier, conventional machine learning techniques can produce inaccurate outcomes, with issue of overfitting. For that reason it become important to develop more effective learning techniques in order to enhance IDS performance. Nowadays cyber-attacks are increasing which leads critical situation for healthcare applications. We require strong protection techniques for both internal and external components to maintain the trustworthiness of IoMT infrastructures.

In today's world, decision-making can become more transparent and reliable by utilising deep learning models and thoroughly examining the effectiveness of various optimizers. It also increases trust in the effectiveness of IDSs and gives researchers a new area to understand and evaluate detection techniques.

IV. BASIC PRELIMINARIES

Optimizers are used with machine learning, which makes a big impact on a model's performance. The choice of optimizer is crucial for deep neural networks (DNNs) because the efficiency of training of the model depends on the parameters of the model. Based on predetermined update rules that direct the refinement of weights during training, optimizers adjust hyperparameters, such as the learning rate. The main goal is to improve the performance based on base of accuracy by iteratively updating network weights in a way that minimises the loss function. In essence, an optimizer aims to reduce training loss and steer the model toward better generalization. This study employs eight different optimization techniques to train the proposed DNN model. A detailed description of each optimizer and its corresponding characteristics is presented in Table 2.

A. SGD

Stochastic Gradient Descent (SGD) is an advanced and faster form of the conventional Gradient Descent algorithm, commonly used for tasks like classification, linear regression, and backpropagation. Its main function is to iteratively minimise the cost function from an initial starting point. The mathematical representation of SGD is provided in equation (1).

$$W_{k+1} = W_k - \alpha \nabla L(W_k; p, q) \qquad (1)$$

B. RMSprop

It uses gradients by using an adaptive learning rate as an alternative of a hyperparameter that variates with time (Eq. (2)).

$$W_{k+1} = W_k - \frac{\alpha}{\sqrt{M_{k+1}} + \varepsilon} \nabla L(W_k; p, q) \qquad (2)$$

$$Where, M_{k+1} = \beta M_k + (1-\beta)\left(\nabla L(W_k; p, q)\right)^2,$$
$$M_k \text{ is the Moving average of the squared gradient},$$
$$\beta\text{: Decay factor (which is close to 1)}$$

C. Adagrad

It automatically adjusts the learning rate for each parameter according to its update history. Parameters decrease more frequently receive smaller learning rates, improving performance and eliminating the need for manual tuning. The mechanism of Adagrad is expressed in equation (3).

$$W_{k+1} = W_k - \frac{\alpha}{\sqrt{GM_{k+1}} + \varepsilon} \nabla L(W_k; p, q) \qquad (3)$$

$$GM_{k+1} = GM_k(\nabla L(W_k; p, q))^2,$$
$$GM_k\text{: Diagonal matrix elements accumulates}$$
$$\text{the squared gradient}$$

D. AdaDelta

AdaDelta is the improved version of Adagrad that remove the necessity for a fixed initial learning rate. It adaptively updates hyper-parameters during training and is more resilient to gradient noise, using a limited window to control the impact of past accumulated gradients. Its mathematical formulation is shown in equation (4).

$$W_{k+1} = W_k + \frac{\sqrt{E[\Delta m^2]_{k-1} + \varepsilon}}{\sqrt{E[n^2]_k + \varepsilon}} \nabla L(W_k; p, q) \qquad (4)$$

E. Adam

Adam, short for Adaptive Moment Estimation, is one of the most extensively used optimization algorithms and shares conceptual similarities with both RMSProp and Adagrad. It utilizes the L2 norm (Euclidean norm) to achieve efficient, adaptable, and fast optimization. As its name suggests, Adam estimates adaptive learning rates for individual network weights by computing the first and second moments of the gradients. The operational mechanism of the Adam optimizer is presented in equation (5).

$$W_{k+1} = W_k - \frac{\alpha}{\sqrt{\frac{V_{k+1}}{1-\beta_2^{K+1}} + \varepsilon}} * \frac{U_{k+1}}{1-\beta_1^{K+1}} \qquad (5)$$

$$where, U_{k+1} = \beta_1 U_k + (1-\beta_1)\nabla L(W_k; p, q),$$
$$V_{k+1} = \beta_2 V_k + (1-\beta_2)\left(\nabla L(W_k; p, q)\right)^2$$

F. Adamax

This algorithm serves as a modified form of Adam. It computes adaptive updates using lower-order moment estimates, making it highly efficient in terms of computation while also requiring minimal memory.

For Adamax Equation (5) updated with

$$V_k\text{: exponentially weighted infinity norm of the gradients}$$
$$V_{k+1} = max\ (\beta_2 . V_k . |\nabla L(W_k; p, q)|$$

G. Nadam

Nadam, formed from the joining of Adam and Nesterov Accelerated Gradient (NAG), integrates Nesterov's momentum into Adam's adaptive moment estimation. By incorporating a look-ahead gradient update, Nadam achieves faster and more stable convergence, particularly in scenarios involving high curvature or noisy gradients. Its learning process benefits from the accelerated reduction in the moving averages of both past and current gradients. The operational formulation of the Nadam optimizer is presented in equation (6).

$$W_{k+1} = W_k - \frac{\alpha}{\sqrt{\frac{V_{k+1}}{1-\beta_2^{K+1}} + \varepsilon}} * \left(\beta_1 \frac{U_{k+1}}{1-\beta_1^{K+1}} + \frac{(1-\beta_1)\nabla L(W_k + \beta_1 U_k; p, q)}{1-\beta_1^{K+1}}\right) \qquad (6)$$

H. AdamW

AdamW optimizer is the improved version of Adam that includes weight decay shown in table.2. It is a regularization process which use a penalty having large weights during the optimization process. AdamW is updated as shown in equation (7).

$$W_{k+1} = W_k - \frac{\alpha}{\sqrt{\frac{V_{k+1}}{1-\beta_2^{K+1}} + \varepsilon}} * \frac{U_{k+1}}{1-\beta_1^{K+1}} + \vartheta W_k \qquad (7)$$

TABLE 2. OPTIMIZERS ABBREVIATION EXPLANATION

Symbol	Description
(\alpha)	Learning parameter
(U_k)	First moment (mean) of gradients
(V_k)	Second moment (variance) of gradients
(\beta_1)	Exponential decay rate for 1st moment
(\beta_2)	Exponential decay rate for 2nd moment
(\vartheta)	Weight decay coefficient

V. METHODOLOGY DISCUSSION

This section describes how to identify and categorise IoMT network vulnerabilities by using deep learning. The system integrates a DNN model and also several optimizers, and offers a thorough examination of every phase, highlighting each one's unique contributions and training time impacts.

A. DNN

A deep neural network (DNN) used multiple neurons organised into layers, functioning as a feed-forward network when processed data moves from input to output [13]. It

typically includes an input layer to receive data, one or more hidden layers to process it, and an output layer to generate results. During training, the network adjusts layer weights to optimize performance. DNNs good in handling tabular and nonlinear data due to their ability to learn optimal mappings between inputs and outputs. Activation functions are useful in dealing with nonlinearity, which allows the network for capturing complex patterns, a property aligned with the universal approximation theorem. Fig 1 illustrates the general structure of a DNN.

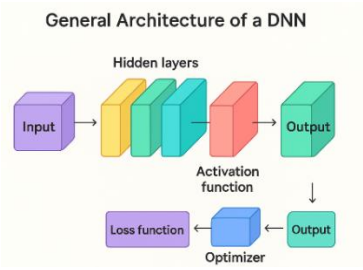


Fig 1. DNN Architecture

B. Proposed Method

In this study, DNN fully connected with four layers was designed and optimized for effective intrusion detection. The initial single-hidden-layer configuration proved inadequate, so multiple architectures were tested, and a two-hidden-layer design delivered the best results, as illustrated in Fig 2. In this fully connected network, each neuron links to every neuron in the next layer. The input vector XXX consists of 66 features (X={x1,x2,...,x66}X = \{x_1, x_2, ..., x_{66}\}X={x1,x2,...,x66}), while the output vector YYY contains six values representing attack classes from (C0 to C5).The DNN architecture includes one hidden layer with 64 and another with 128 neurons, respectively, both using the ReLU [14], and a softmax output layer with six neurons. Mathematically it is expressed in equation (8).

$$HL_i(X) = f_{activation}(W_i X + b_i) \qquad (8)$$

$f_{activation}(.)$ is denoting the non-linear activation function. Similarly, W_i symbolizes the weight factor and b_i represents the bias factor.

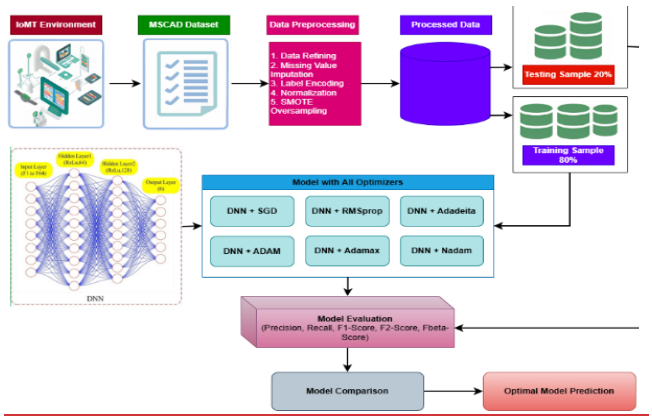


Fig 2. Model Diagram of the proposed framework

The output nodes and hidden nodes are subjected to the softmax activation function [15] and the ReLu activation function, respectively. In equations (9) and (10), the ReLu and softmax are considered.

$$f_{activation(ReLu)} = \max(0, X) \qquad (9)$$

$$f_{activation(softmax)} = \frac{e^{x_i}}{\sum_{k-1}^{n} e^{x_k}} \qquad (10)$$

Here the performance is to be improved by suitably decreasing the loss value.

VI. RESULT ANALYSIS

This section provides a broad view on MSCAD dataset, which is used inside the experiment to train the model using multiple optimizers, and it includes processes such as preparation of the dataset, experimental setup and at the end a comprehensive analysis of results that we got at the end for evaluation of identification of attacks within the IoMT environment.

A. Dataset Overview

Here we used the MSCAD dataset and the overview can find in the Table 3. This dataset was downloaded from IEEE DataPort [16] and this dataset consists of 66 feature columns, and one more column is there "Label". On study the dataset, which is consists of two primary attack types that are carried out in stages such as volume-based DDoS attacks and brute-force password attacks. In the brute-force scenario, an attacker scans ports, copies web pages offline using HTTrack Website Copier, and then targets hosts with a DDoS attack. A traffic-based DDoS using Radware, an HTTP Slowloris attack, and Full, SYN, FIN, and UDP port scans are all part of the volume-based DDoS attack. With 128,799 examples, the dataset offers a thorough standard for intrusion detection studies.

TABLE 3. DATASET DESCRIPTION

Feature		***Attack Class Description***					
Features	Target Vectors	Brute Force	HTTP DDoS	ICMP Flood	Normal	Port Scan	Web Crawling
66	1	88502	641	45	28502	11081	28

B. Experimental Setup

Language	Python v 3.9
Framework	Keras, Tensorflow
Hardware	HP
OS	Windows 11 Home
Processor	Intel Core i5 & GPU

C. Data Preparation

MSCAD dataset was preprocessed through five main steps on data, so that the training of the model will start. First, pandas to clean up the data by removing any duplicates and inconsistent entries to make sure the data was reliable. Then deal with any missing values by dropping records that had NaNs. After that, converted categorical fields, like the 'Label' column, into numerical values using label encoding. To help with training stability and convergence, In next step of preprocessing normalized all feature values with the min–max scaling technique. Since the dataset had a significant class imbalance, so implemented the SMOTE oversampling method with the 'auto' strategy to even things out. 80% training and 20% testing samples are considered in this experiment, ensuring a random sampling for a fair separation.

D. Evaluation Metrics

The efficiency of the algorithm was evaluated by using five standard performance metrics shown in Table 4. The performance of each models are compared for best analysis of the result.

E. Experimental Result Analysis

This section presents the influence of different optimiser algorithms on the performance of the DNN model using the MSCAD dataset. This study also comparations the proposed DNN-based attack detection framework with several traditional ML techniques, including LR, DT and ensemble models such as RF, Bagging, GBoost and AdaBoost. The qualified performance of these ML models against the proposed DNN–Adam architecture is summarized in Table 4, while Table 5 provides the experimental results demonstrating the variance on the optimizer's performance on the DNN model. In table 4, PF1 is the precision, PF2 is the Recall, PF3 is the F1 score, PF4 is the F2 score and Pf5 is Fβ score.

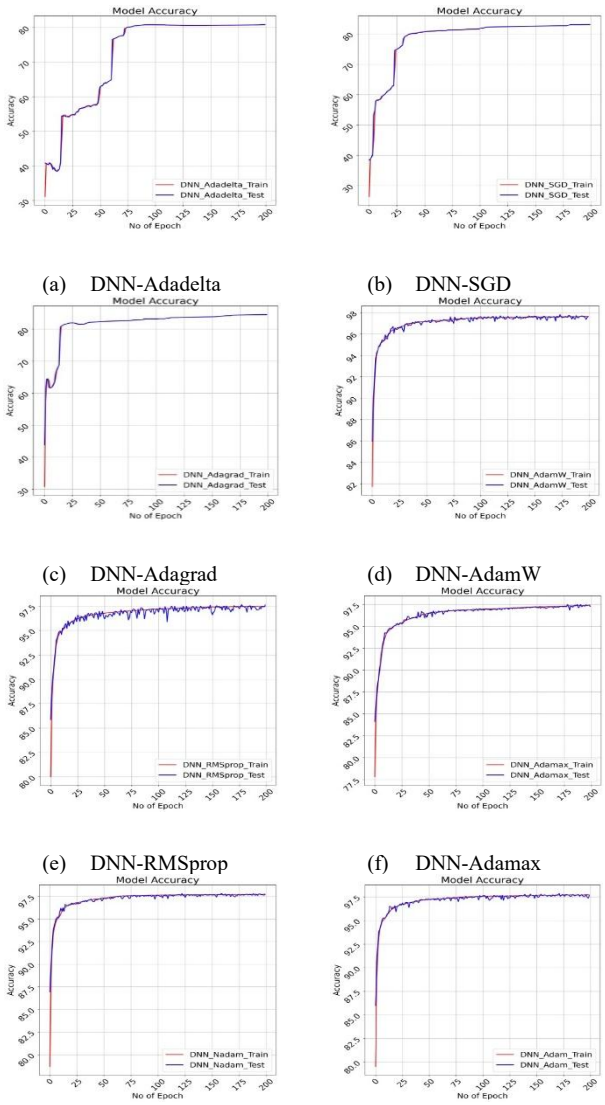
TABLE 4. THREE ML MODEL COMPARED TO DNN-ADAM

Name of the Model	PF1	PF2	PF3	PF4	PF5
DT	0.78882	0.79261	0.78924	0.78923	0.7783
LR	0.63673	0.59624	0.61154	0.61152	0.6014
RF	0.98222	0.92445	0.94993	0.94973	0.9422
Proposed model	0.97751	0.97731	0.97722	0.97712	0.9670

TABLE 5. RESULTS ANALYSIS WITH DIFFERENT OPTIMIZERS

Classification Model	PF1	PF2	PF3	PF4	PF5	GPUTime Elapsed (Minutes)
DNN-Adadelta	0.832	0.80454	0.803	0.803	0.803	12.054
DNN-SGD	0.852	0.83303	0.83294	0.832	0.832	10.333
DNN-Adagrad	0.863	0.84412	0.84464	0.844	0.844	12.224
DNN-AdamW	0.975	0.97555	0.975	0.975	0.975	11.466
DNN-RMSprop	0.976	0.9760	0.9752	0.97572	0.97584	12.202
DNN-Adamax	0.9768	0.976703	0.9767	0.97663	0.97675	9.5603
DNN-Nadam	0.9773	0.977202	0.977	0.97712	0.97723	10.4112
Proposed Model	0.9775	0.97735	0.97722	0.97719	0.97726	9.4056

However, traditional ML models, Adaboost demonstrates the highest performance, achieving an F1-score of 0.9318 and an overall accuracy of 0.9204. These accuracy levels still fall noticeably short of the effectiveness achieved by the DNN-Adam model. The results clearly show that the integration of an appropriate optimizer significantly enhances the performance of the DL model. As evidenced in Table 5, the proposed DNN-Adam architecture surpasses all other evaluated models and proved to be the best performing model.



(a) DNN-Adadelta (b) DNN-SGD

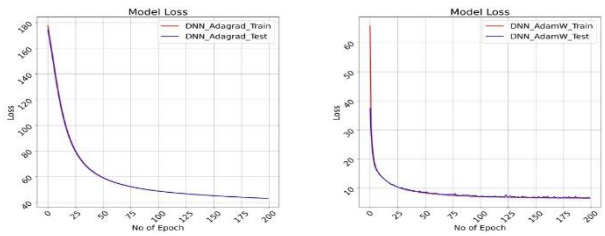
(c) DNN-Adagrad (d) DNN-AdamW

(e) DNN-RMSprop (f) DNN-Adamax

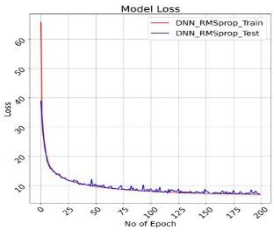
(g) DNN-Nadam (h) Proposed DNN-Adam

Fig 3. Validation Accuracy Analysis of all models

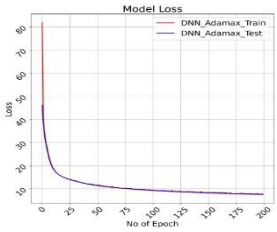
The graphs in fig 3, shows the performance of various optimizers applied to the DNN model for 200 epochs with respect to validation accuracy. The graph, it shows DNN – Adadelta setup performs the worst result as compare to others. In other hand optimizers Adamax, Nadam, and Adam perform noticeably better. DNN – Adam experiment setup is the most successful optimizer, which consistently achieves the highest accuracy and the fastest convergence.



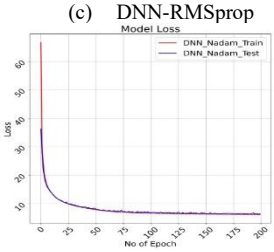
(a) DNN-Adagrad (b) DNN-AdamW



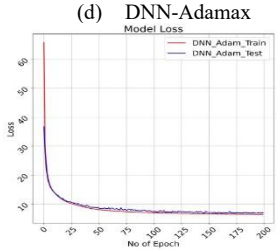
(c) DNN-RMSprop



(d) DNN-Adamax



(e) DNN-Nadam



(f) Proposed DNN-Adam

Fig 4. Validation Loss Analysis of all models

From the above fig 4 shows the losses of different optimizers which influenced the performance of DNN models by the nature of datset and optimizer's characteristics. These differ in performance arises from the distinct ways by which the optimizer affects the model performance such as ability to learn meaningful internal representations. After all analysis, it is found that DNN – Adam setup achives the lowest loss as compare to others, further confirming its robustness and effectiveness. From the result of DNN – Adam algorithm found that it achieves a good score in F1-Score having less training and execution time as compared to all other optimizers.

VII. Conclusion

Effective identifying unauthorized entries is critical within an IoMT framework. On the based of optimizer integration with Deep Neural Networks (DNN), are employed for IDS in this investigation. To assess the performance of diverse Deep Neural Network (DNN) optimizers, experiments were executed with the ensuing parameters. The performance has been achieved with a setting of no. of epochs 200, 66 neurons at the input layer, 1st hidden layer encompassing 64 units and so on. The model was assessed on the Medical Smart City Attack Dataset (MSCAD), which encompasses six distinct categories of attacks. In addition to evaluating the intrusion detection, this work is taken several other issues and domains, which include difference detection and disease identification. As experimental result shows strong accuracy, the model requires considerable computational resources, as is common with DNN-based systems. Future work will explore automated hyper-parameter optimization and advanced feature selection techniques to reduce computational cost and further improve performance.

References

[1] Litoussi, Mohamed, et al. "IoT security: challenges and countermeasures." Procedia Computer Science 177 (2020): 503-508. Doi: 10.1016/j.procs.2020.10.069.

[2] Razzaq, Mirza Abdur, et al. "Security issues in the Internet of Things (IoT): A comprehensive study." International Journal of Advanced Computer Science and Applications 8.6 (2017). Doi: 10.14569/ijacsa.2017.080650.

[3] Nayak, Janmenjoy, Pooja Puspita Priyadarshani, and Pandit Byomakesha Dash. "Improved perturbation based hybrid firefly algorithm and long short-term memory based intelligent security model for IoT network intrusion detection." *Computers and Electrical Engineering* 121 (2025): 109926.

[4] Subhan, Fazli, et al. "AI-enabled wearable medical internet of things in healthcare system: A survey." Applied Sciences 13.3 (2023): 1394. Doi: 10.3390/app13031394.

[5] Aminizadeh, Sarina, et al. "The applications of machine learning techniques in medical data processing based on distributed computing and the Internet of Things." Computer Methods and Programs in Biomedicine (2023): 107745. Doi: 10.1016/j.cmpb.2023.107745.

[6] Luo, Gang. "Automatically explaining machine learning prediction results: a demonstration on type 2 diabetes risk prediction." Health information science and systems 4 (2016): 1-9. Doi: 10.1186/s13755-016-0015-4.

[7] Reddy, Dukka Karun Kumar, Janmenjoy Nayak, H. S. Behera, Vimal Shanmuganathan, Wattana Viriyasitavat, and Gaurav Dhiman. "A Systematic Literature Review on Swarm Intelligence Based Intrusion Detection System: Past, Present and Future." *Archives of Computational Methods in Engineering* 31, no. 5 (2024).

[8] Newaz, AKM Iqtidar, et al. "Healthguard: A machine learning-based security framework for smart healthcare systems." 2019 sixth international conference on social networks analysis, management and security (SNAMS). IEEE, 2019. Doi: 10.1109/SNAMS.2019.8931716.

[9] Liu, Jinxin, Burak Kantarci, and Carlisle Adams. "Machine learning-driven intrusion detection for Contiki-NG-based IoT networks exposed to NSL-KDD dataset." Proceedings of the 2nd ACM workshop on wireless security and machine learning. 2020. Doi: 10.1145/3395352.3402621.

[10] Nayak, Janmenjoy, Saroj K. Meher, Alireza Souri, Bighnaraj Naik, and S. Vimal. "Extreme learning machine and bayesian optimization-driven intelligent framework for IoMT cyber-attack detection." *The Journal of Supercomputing* 78, no. 13 (2022): 14866-14891.

[11] Ahmad, Muhammad, et al. "Intrusion detection in internet of things using supervised machine learning based on application and transport layer features using UNSW-NB15 data-set." EURASIP Journal on Wireless Communications and Networking 2021.1 (2021): 1-23. Doi: 10.1186/s13638-021-01893-8.

[12] Iwendi, Celestine, et al. "Security of things intrusion detection system for smart healthcare." Electronics 10.12 (2021): 1375. Doi: 10.3390/electronics10121375.

[13] Mishra, Debasmita, Bighnaraj Naik, Janmenjoy Nayak, Alireza Souri, Pandit Byomakesha Dash, and S. Vimal. "Light gradient boosting machine with optimized hyperparameters for identification of malicious access in IoT network." *Digital Communications and Networks* 9, no. 1 (2023): 125-137.

[14] Agarap, Abien Fred. "Deep learning using rectified linear units (relu)." arXiv preprint arXiv:1803.08375 (2018). Doi: 10.48550/arXiv.1803.08375.

[15] Nwankpa, Chigozie, et al. "Activation functions: Comparison of trends in practice and research for deep learning." arXiv preprint arXiv:1811.03378 (2018). Doi: 10.48550/arXiv.1811.03378.

[16] Mohammad Almseidin, Jamil Al-Sawwa, Mouhammd Alkasassbeh, "Multi-Step Cyber-Attack Dataset (MSCAD for Intrusion Detection)", IEEE Dataport, June 18, 2022, doi:10.21227/phr0-e264